Statistics 212: Lecture 13 (3/24 2025)

Prohorov's Theorem and Paley-Wiener Integral

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1 Finite Variation / Convergence

Last time, we saw Donsker's Theorem. If $X_n = \pm 1$ are i.i.d., then

$$S_n = \sum_{i=1}^{n} X_i$$

converges to Brownian Motion under rescaling.

An alternative perspective: in general, for a (S, d) complete separable metric space with a sequence of random variables $X_1, X_2, \ldots$, we want conditions under which $X_i \xrightarrow{d} X$.

Theorem 1. *Suppose:*

1. *Every subsequence $\{X_{n_k}\}$ has a convergent sub-subsequence in distribution (equivalent to tightness).*
2. *If a subsequence $\{X_{m_k}\}$ converges in distribution, then the limit is X.*

Then $X_i \xrightarrow{d} X$.

Intuition: Suppose $S = \mathbb{R}$. If $X_i \in \mathbb{R}$ are deterministic and bounded, and there is only 1 *subsequential limit point, then X_i must converge to that limit.*

Proof. Suppose $X_i \overset{d}{\not\to} X$. Then there is a bounded, continuous function $h : S \to \mathbb{R}$ such that

$$\int h \, d\mu_i \not\to \int h \, d\mu.$$

Hence we can extract a subsequence N_k and $\epsilon > 0$ with

$$\left| \int h \, d\mu_{N_k} - \int h \, d\mu \right| \geq \epsilon, \quad \forall k.$$

By (1), μ_{N_k} converges in distribution to some $\tilde{\mu}$. By (2), $\tilde{\mu} = \mu$. This contradiction completes the proof. $\square$

Example 1. *Consider $S = C([0,1])$. To show part (2) of Theorem 1, it suffices to show*

$$\big(X_n(t_1), \ldots, X_n(t_k)\big) \xrightarrow{d} \big(X(t_1), \ldots, X(t_k)\big)$$

for all $0 \le t_1 < \cdots < t_k \le 1$, which essentially amounts to showing a multidimensional CLT. The reason this suffices is that, as we have seen, any random continuous path with the finite-dimensional distributions of Brownian motion is *Brownian motion.*

Definition 1 (Tightness)**.** *We call $\{X_n : n \ge 1\}$* tight *if, for all $\epsilon > 0$, there exists a compact $K_\epsilon \subseteq S$ such that*

$$\sup_{n \ge 1} P\big[X_n \in K_\epsilon\big] \;\ge\; 1 - \epsilon.$$

Prokhorov's Theorem tells us that condition (1) in Theorem 1 is equivalent to tightness. In $C([0,1])$, we typically check equicontinuity (Arzelà–Ascoli).

Question 1. *Why is a single random variable on μ a tight family?*

Answer sketch: In a complete metric space, compactness is equivalent to "closed + totally bounded." (A subset of a metric space is totally bounded if for any $\epsilon > 0$, it can be covered by finitely many balls of radius ϵ.) One shows that for every $\epsilon > 0$, there is a compact set K_ϵ covering all but ϵ of the probability mass. This is done by choosing countable dense sets, finite covers, and a diagonal argument.

Why tightness implies (1). We take $\epsilon_j = 1/2^j$ and partition K_{ϵ_j} into sets of diameter $\le \epsilon_j$. Then we construct subsequences so that μ_{N_i} converges on each partition. By a diagonal argument, we get convergence on all levels. By the Portmanteau Theorem, any bounded, Lipschitz h then has $\int h \, d\mu_{N_i}$ convergent.

Corollary 1. *We can metrize convergence in distribution by*

$$d(\mu, \nu) \;=\; \sup_{h : S \to \mathbb{R}} \Big| \int h \, d\mu - \int h \, d\nu \Big|,$$

where the supremum is over bounded Lipschitz h. Two consequences of Prokhorov's Theorem:

- *If S is compact,* $\mathrm{Prob}(S)$ *is compact.*
- *If S is complete and separable,* $\mathrm{Prob}(S)$ *is complete and separable.*

Example 2 (Optimal Transport)**.** *Given $\mu, \nu \in \mathrm{Prob}(S)$, we look for a coupling $\alpha \in \mathrm{Prob}(S \times S)$ with marginals μ, ν. Then*

$$\min_{\alpha : (\alpha_1, \alpha_2) = (\mu, \nu)} \int h(x_1, x_2) \, d\alpha(x_1, x_2)$$

for some cost h. If h is bounded and continuous, an optimal α_* exists.

2 Stochastic Integration

We want to define

$$\int_0^s a_t \, dB_t,$$

where B_t is Brownian Motion. In Itô diffusions, we often write

$$dX_t = a_t \, dB_t + b_t \, dt$$

and equivalently

$$X_t \;=\; \int_0^t a_s \, dB_s \;+\; \int_0^t b_s \, ds.$$

In general, a_s and b_s can depend on B_s or even the full history $B_{[0,s]}$. Integrating ds is just calculus, but integrating dB_s is a new operation! Today we will start with the case that a_s is a deterministic function of time.

Definition 2 (Paley–Wiener Integration)**.** *Suppose a is a deterministic function. First assume a is piecewise constant:*

$$a(s) = A_k \quad \textit{on } [t_{k-1}, t_k].$$

Then

$$\int_0^1 a(s) \, dB_s \;=\; \sum_k A_k \left(B_{t_k} - B_{t_{k-1}}\right).$$

If $\|a - \tilde{a}\|_{L^2} = D$, then

$$\left\| \int_0^1 a(t) \, dB_t \;-\; \int_0^1 \tilde{a}(t) \, dB_t \right\|_{L^2} \;=\; D.$$

Hence for any $a \in L^2([0,1])$, we define $\int_0^1 a(s) \, dB_s$ as the L^2-limit of $\int_0^1 a^{(k)}(s) \, dB_s$ for piecewise constant $a^{(k)} \xrightarrow{L^2} a$.

Proof. Since L^2 is an inner-product space, the key is to show

$$E\Big[\int_0^1 a(t) \, dB_t \;\times\; \int_0^1 \tilde{a}(t) \, dB_t\Big] \;=\; \int_0^1 a(t) \, \tilde{a}(t) \, dt.$$

Both sides are bilinear. It suffices to check it for a and $\tilde{a}$ as indicators of intervals $[I, J]$ and $[\tilde{I}, \tilde{J}]$, then extend by linearity. This is computed by casework on how the two intervals overlap (we did one case in class).

Having done so, bilinearity implies the above identity for any piecewise constant functions $a, \tilde{a}$. Since piecewise constant functions are dense in $L^2([0,1])$, using continuity yields an isometry from $L^2([0,1]) \to L^2(\mathcal{F})$, where $\mathcal{F}$ is the sigma-field generated by $B_{[0,1]}$. This is the Paley-Wiener integral. $\square$